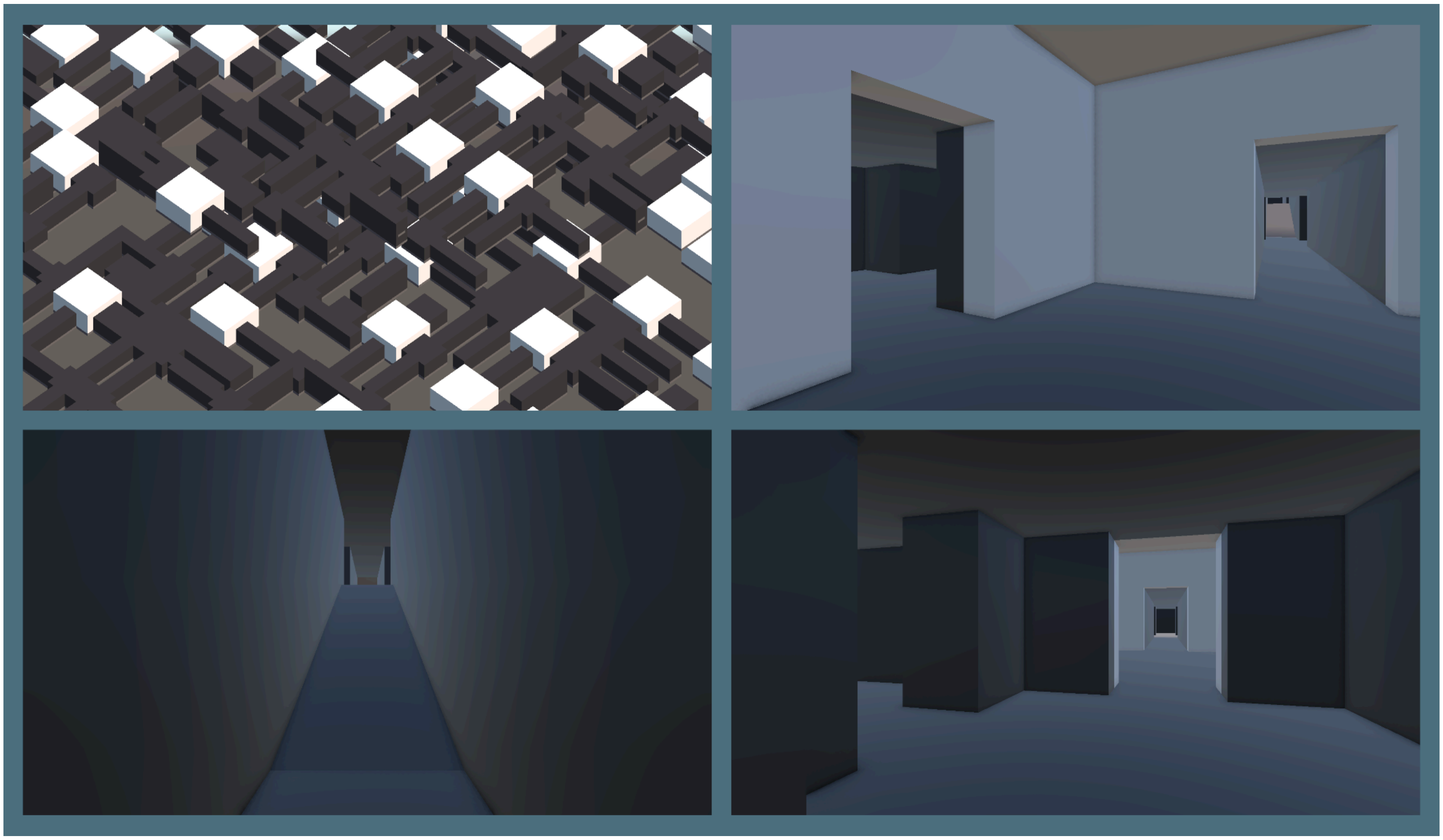
<u>Dungeon Generation</u>

by A Door To gaming



This Dungeon Was Generated With The Sample Generation Type With A Higher Room Count.

Dungeon Generation:

Put a Dungeon Generation component on any object to turn it into a Dungeon Generator.

Options:

1. Generate On Start
    - If set to true it generates a dungeon once the scene it's in has been loaded.

2. Gen Type
    - The rules of the generator.

3. Generation Level
    - What level rooms are allowed to generate.

4. On Generation Start()
    - Any function here will execute once the generation has started.

5. On Generation Complete()
    - Any function here will execute once the generation has ended.

6. Log Mode
    - None: Will only log if an error occurs.
    - Normal: Logs once the generator is starting and has completed the dungeon generation.
    - Debug: Logs everything for debugging.

Dungeon Generation Type

A Scriptable Object that controls the rules of the generator.

Options:

1. Min/Max Room Count
   - A random number between these values will control how many rooms the dungeon will have.

2. First Group
   - What room group to generate from first. (Leave blank for a random group)

3. Rooms
   a. Dungeon Room
      - A reference to the room it will generate.

   b. Weight
      - The chance of this room being picked to get generated (The higher it is the more chance of it being picked.).

   c. Level
      - What minimum generation level/difficulty is required for this room to generate.

4. Default Dead End
   - A reference to the dead end used to block off any unused sections if it's not being overridden.

You can create a new Dungeon Generation Type in <u>"Create>Dungeon Generation>Dungeon Generation Type"</u>

<u>Dungeon Generation Room</u>

The room component for any room prefab. Has a trigger box with an offset and size variable. This detects if the room is overlapping with any other room for a successful generation.

Options:

1. Room Offset
    - The offset of the overlap detection trigger.
2. Room Size
    - The size of the overlap detection trigger.
3. The group this room is a part of. This allows for more customisation.

<u>Dungeon Generation Node</u>

A node where a room should spawn and its spawning offset.

Options:

1. Chance To Generate From Different Groups
    - The change to generate a room from a different group then itself. (0 makes it always generate from the same group) (100 makes it always generate from a different group).

2. Override Dead End
    - Overrides the default dead end if a dead end for this node should spawn here.

Sample Dungeon

This is a good example of how the generation system works.

Contains:

- 2x Basic Rooms
- 2x Hallways
- 1x Big Room
- Configured Dungeon Type
- Rooms with different levels and weights
- Rooms with different groups and node counts.

I recommend checking out the sample before making your own if this is the first time using the system.

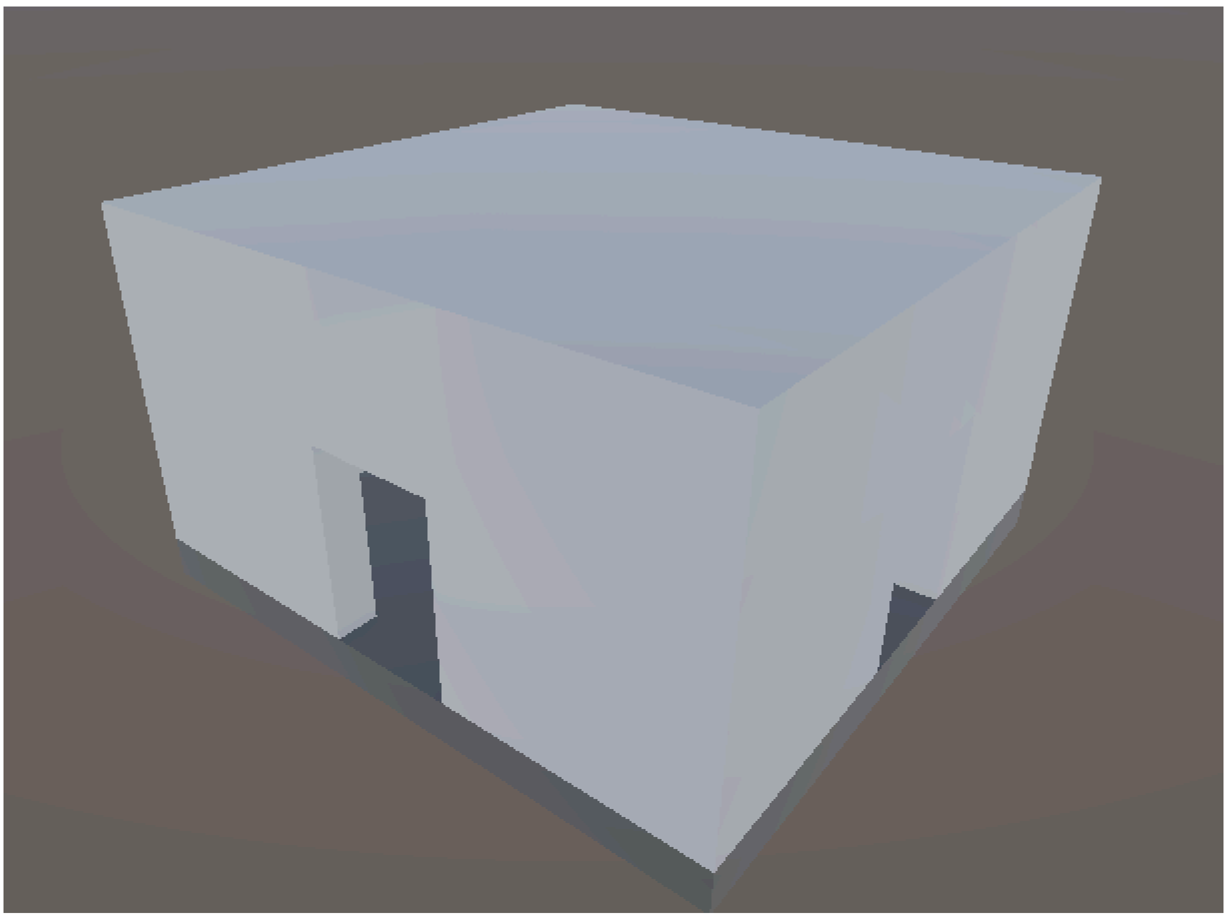
How To Create A Room

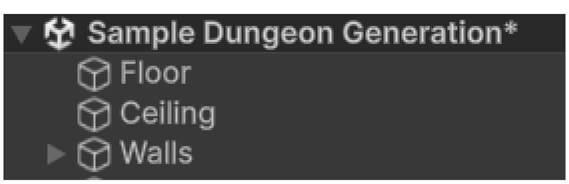
A step by step tutorial on how to create a room.

You can copy and paste a room from the sample dungeon or follow each step.

Step 1:

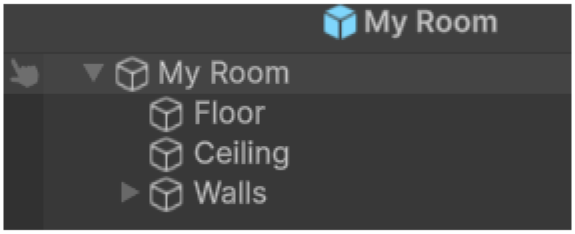
Start by creating a room layout, what the room is going to look like.

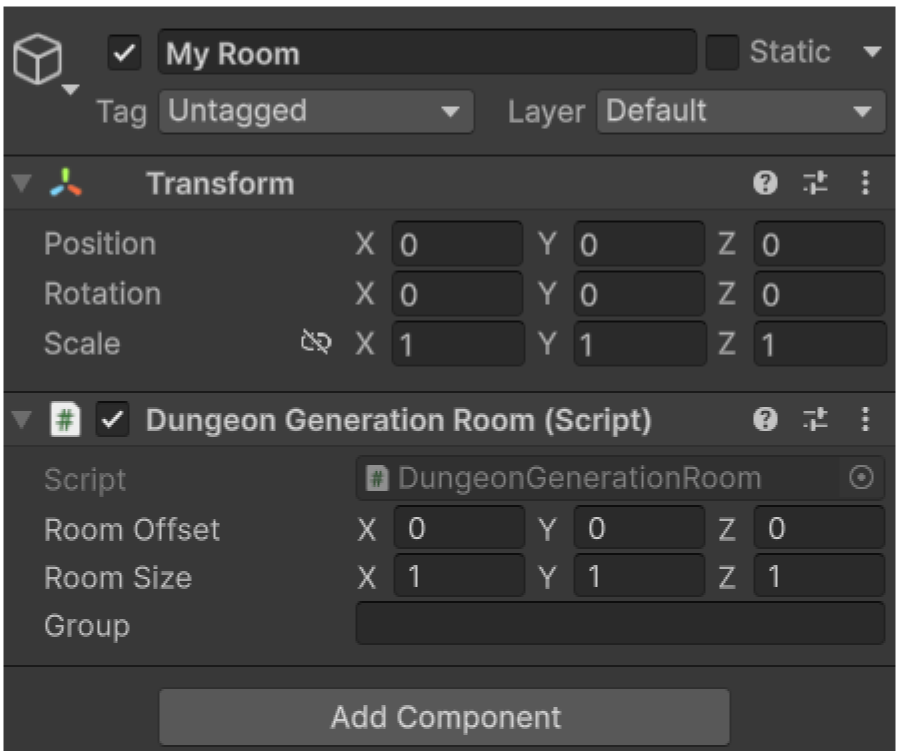




Step 2:

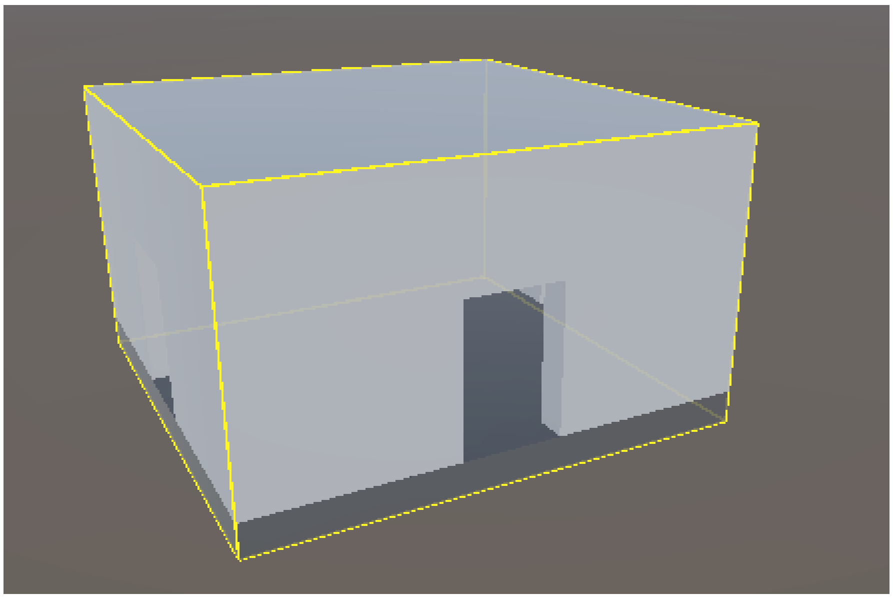
Put the room into an empty game object with the Dungeon Generation Room script attached. Now make that empty game object into a prefab.

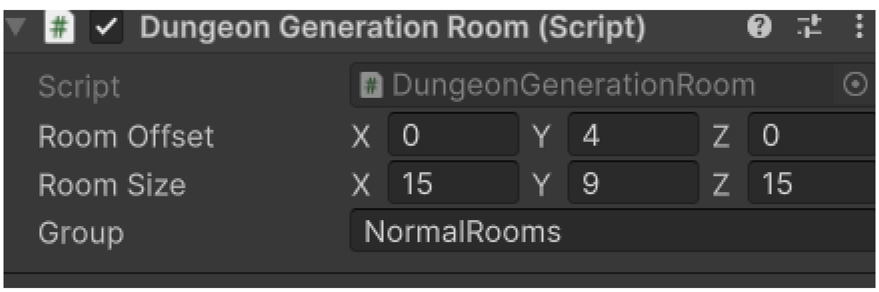




Step 3:

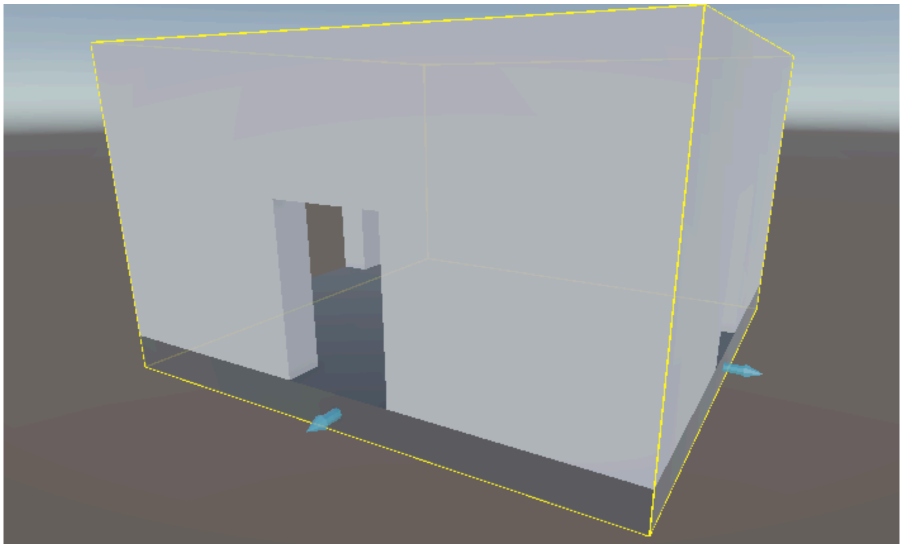
Configure the overlap detection trigger of the room. It is represented by a yellow square. If you do not see a yellow square make sure that the size of the room is not 0 and that Gizmos are enabled. Then put in what Group you want your room to be a part of.

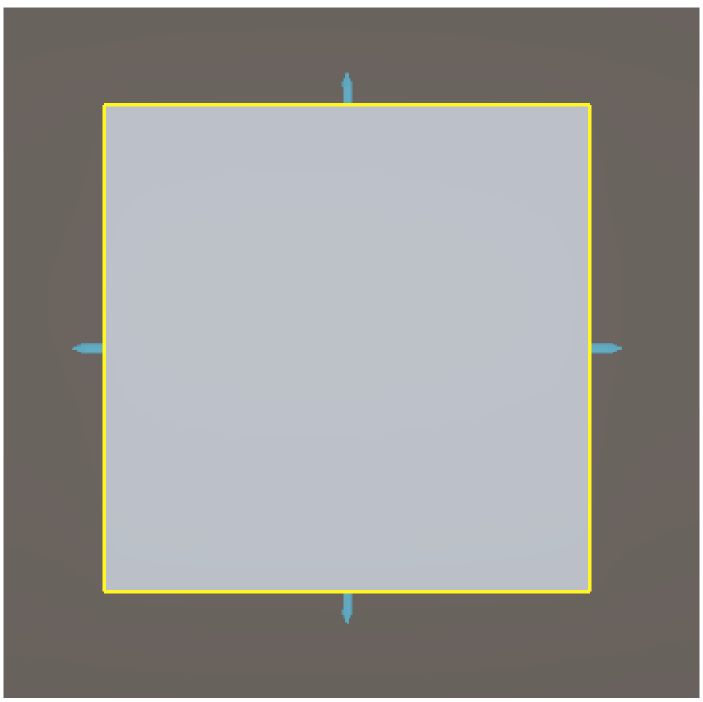


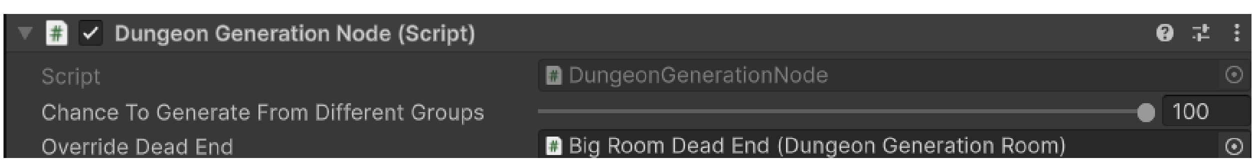


Step 4:

Place down nodes to make the room be able to connect to other rooms and spawn more. A room has to have at least one node. (There is already a node Prefab ready to be used). Then configure them.







<u>Step 5:</u>

Put the room into your desired Dungeon Generation Type then change its weight and level.